

# Beyond Accuracy:

## A Comparative Analysis of Probabilistic and Non-Probabilistic Machine Learning Models for Stroke Risk Screening

Abia TechRise Cohort 3 — AI/ML Track  
GW4 Assignment — Classification Model Comparison

Dataset: Health Stroke Dataset  
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# 1. Introduction

Stroke does not announce itself in advance, but its risk factors usually do, age, hypertension, heart disease, average glucose level, and smoking history are all data points that health systems already record routinely. The central question this study explores is whether that routinely collected data can be used to flag risk early enough to matter.

This is a common challenge in health screening: the condition being predicted (stroke) is rare relative to the overall population, so a model can look highly accurate on paper while still missing most of the people who actually need help. The guiding question of this study is therefore not simply “which model scores highest overall,” but “which model actually catches at-risk patients, and how confidently.”

## Problem Statement

Stroke prediction from routinely collected clinical and demographic data is a highly imbalanced classification problem: the overwhelming majority of patients in the dataset never experience a stroke. Standard classifiers trained on such data tend to default to predicting the majority (“no stroke”) class, which produces misleadingly high accuracy while failing at the task that actually matters clinically catching the rare positive cases.

## Objectives

- Prepare the Health Stroke Dataset (cleaning, encoding, scaling) for fair comparison across models.
- Build four classifiers spanning two philosophies: Logistic Regression and Naïve Bayes (probability-based) versus Support Vector Machine and K-Nearest Neighbours (boundary-based).
- Tune and justify each model's key hyperparameters.
- Compare all four models using accuracy, precision, recall, and F1-score, weighting recall heavily given the clinical cost of a missed stroke case.
- Use ROC curves and AUC to assess the probabilistic models' confidence, not just their verdicts.
- Recommend a model based on both quantitative performance and real-world clinical trustworthiness.

# 2. Data Preprocessing

The Health Stroke Dataset was imported using pandas and inspected with **df.info()** to check data types and completeness. The following preprocessing steps were then applied:

**Missing values.** The **bmi** column was the only field with missing values; these were imputed with the column median rather than dropped, to preserve sample size.

**Duplicates & cardinality.** The dataset was checked for duplicate rows (none found) and the cardinality of each categorical column was reviewed to understand the encoding required.

**Irrelevant / low-signal fields.** The **id** column was dropped as it is a patient identifier with no predictive value. The single row with **gender = 'Other'** was removed, as one observation is too few to learn a reliable pattern from.

**Encoding categorical variables.** Nominal categorical variables (gender, ever\_married, work\_type, Residence\_type, smoking\_status) were one-hot encoded using **pd.get\_dummies()**, since none of these categories carry a natural order.

**Train/test split.** The cleaned, encoded data was split 80:20 into training and test sets using **train\_test\_split**, with the target variable being **stroke**.

**Feature scaling.** The continuous numerical features — **age**, **avg\_glucose\_level**, and **bmi** — were standardised using **StandardScaler**, which is important for distance- and margin-based models such as KNN and SVM, and also benefits the convergence of Logistic Regression.

### 3. Model Development & Evaluation

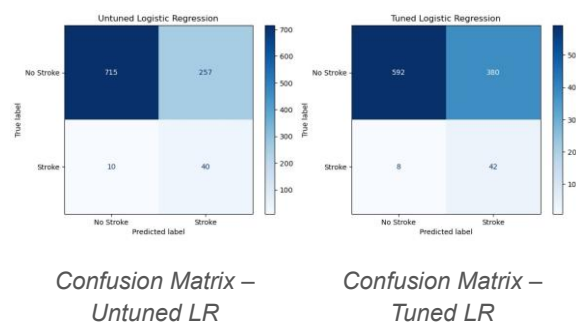
Four classification models were developed: two probabilistic models (Logistic Regression, Naïve Bayes) and two non-probabilistic / boundary-based models (Support Vector Machine, K-Nearest Neighbours). Each model was first trained with default settings, then tuned via **GridSearchCV** (5-fold cross-validation, scored on recall chosen because missing a real stroke case is far costlier than a false alarm in a screening context) and re-evaluated on the held-out test set.

#### 3.1 Logistic Regression (Probabilistic)

A baseline model was trained with **class\_weight='balanced'** to counter the class imbalance, **max\_iter=1000**, and **random\_state=42**. Hyperparameters were then tuned over a grid of **C** values (0.001–10) and **penalty** type (L1/L2, with the liblinear solver), optimising for recall via 5-fold cross-validation. The best combination found was **C=0.005**, **penalty='l1'**, **solver='liblinear'** — a small C imposes strong L1 regularisation, which shrinks less-informative feature coefficients toward zero and biases the decision boundary to flag more borderline cases as at-risk.

| Model        | Parameters        | Accuracy | Precision | Recall | F1-score |
|--------------|-------------------|----------|-----------|--------|----------|
| LR (Untuned) | C=1 (default), L2 | 0.739    | 0.135     | 0.80   | 0.231    |
| LR (Tuned)   | C=0.005, L1       | 0.620    | 0.100     | 0.84   | 0.178    |

Tuning traded overall balance for recall: the tuned model caught more real stroke cases (recall rose from 0.80 to 0.84), but at the cost of accuracy, precision, and F1. Since missed cases matter more than false alarms in this screening context, the tuned model was judged the better fit despite its weaker aggregate statistics.



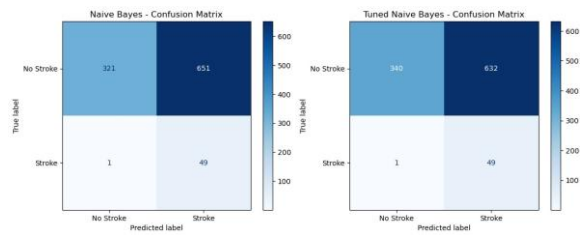
#### 3.2 Naïve Bayes (Probabilistic)

A Gaussian Naïve Bayes classifier was trained with default settings, then tuned over **var\_smoothing** (100 values log-spaced between 1 and 1e-9), again optimising for recall via 5-fold cross-validation. The best value found was **var\_smoothing  $\approx 3.51\text{e-}09$** .

| Model        | Parameters                   | Accuracy | Precision | Recall | F1-score |
|--------------|------------------------------|----------|-----------|--------|----------|
| NB (Untuned) | var_smoothing=1e-9 (default) | 0.362    | 0.070     | 0.98   | 0.131    |
| NB (Tuned)   | var_smoothing=3.51e-09       | 0.381    | 0.072     | 0.98   | 0.134    |

Tuning barely changed anything. Both the untuned and tuned models flagged almost every patient as “at risk,” catching 98% of real stroke cases — but only because they over-predicted stroke for nearly everyone, not because either model was actually discriminating well (accuracy of only 36–38%). This happened because the grid search was scored on recall alone: since predicting “stroke” for almost everyone already achieves near-perfect recall, the search had little reason to move away

from that behaviour. In short, tuning **var\_smoothing** could not meaningfully fix Naïve Bayes's core limitation on this imbalanced dataset.



Confusion Matrix –  
Untuned NB

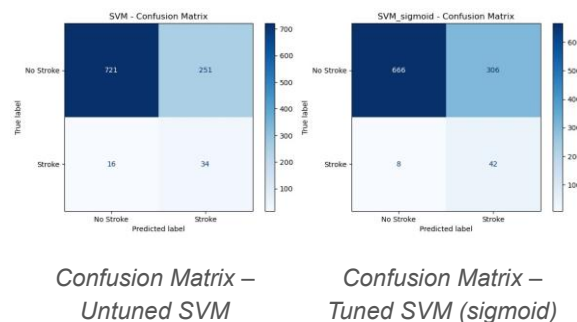
Confusion Matrix –  
Tuned NB

### 3.3 Support Vector Machine (Non-Probabilistic)

A baseline SVM was trained with the default RBF kernel and **class\_weight='balanced'**. Hyperparameters were then tuned over kernel type (linear, RBF, polynomial, sigmoid), **C** (0.1, 1, 10), and **gamma** (scale, auto), again optimising for recall via 5-fold cross-validation. The best combination found was **C=0.1, gamma='auto', kernel='sigmoid'**.

| Model         | Parameters                        | Accuracy | Precision | Recall | F1-score |
|---------------|-----------------------------------|----------|-----------|--------|----------|
| SVM (Untuned) | C=1 (default), RBF                | 0.739    | 0.119     | 0.68   | 0.203    |
| SVM (Tuned)   | C=0.1, gamma=auto, kernel=sigmoid | 0.693    | 0.121     | 0.84   | 0.211    |

Tuning improved SVM cleanly: recall rose from 0.68 to 0.84 (8 more real stroke cases caught), while precision and F1 also improved slightly. Accuracy dropped somewhat, which is the expected trade-off of flagging more patients as at-risk.

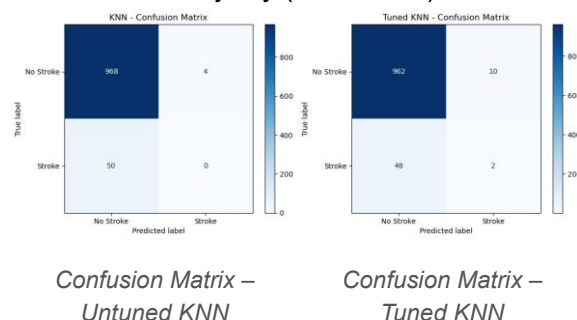


### 3.4 K-Nearest Neighbours (Non-Probabilistic)

A baseline KNN model was trained with scikit-learn's defaults ( $k=5$ , uniform weights, Minkowski distance). Hyperparameters were tuned over **n\_neighbors** (3–21), **weights** (uniform/distance), and **metric** (Euclidean/Manhattan), optimising for recall via 5-fold cross-validation. The best combination found was **metric='manhattan', n\_neighbors=3, weights='distance'**.

| Model         | Parameters                          | Accuracy | Precision | Recall | F1-score |
|---------------|-------------------------------------|----------|-----------|--------|----------|
| KNN (Untuned) | $k=5$ (default), uniform, minkowski | 0.947    | 0.000     | 0.00   | 0.000    |
| KNN (Tuned)   | manhattan, $k=3$ , distance         | 0.943    | 0.167     | 0.04   | 0.065    |

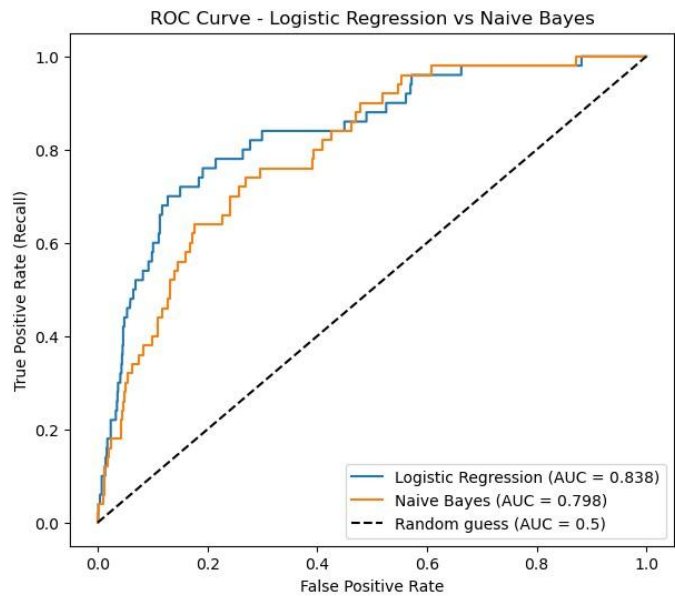
Tuning barely improved KNN — recall only rose from 0.00 to 0.04 (2 of 50 real cases caught). Unlike Logistic Regression and SVM, KNN has no built-in mechanism to account for class imbalance (it has no **class\_weight** parameter), so no combination of  $k$ , weighting, or distance metric could meaningfully correct its bias toward the majority (“no stroke”) class.



### 3.5 ROC Curves & AUC (Probabilistic Models)

Because Logistic Regression and Naïve Bayes output class probabilities, their discriminative power was additionally assessed using ROC curves and the Area Under the ROC Curve (AUC), which

measure how well each model ranks true stroke cases above non-stroke cases across every possible decision threshold — independent of the specific threshold used for the classification tables above.



ROC Curve – Tuned Logistic Regression vs. Tuned Naïve Bayes

| Model                       | ROC-AUC |
|-----------------------------|---------|
| Logistic Regression (Tuned) | 0.838   |
| Naïve Bayes (Tuned)         | 0.798   |

Logistic Regression achieves the higher AUC (0.838 vs. 0.798), meaning it separates stroke from non-stroke patients more reliably across all thresholds, even though Naïve Bayes shows a slightly higher recall at its default 0.5 threshold. This confirms that Naïve Bayes’s high recall is a symptom of systematic over-prediction rather than genuinely stronger discrimination — its ranking of patients by risk is less trustworthy than Logistic Regression’s.

#### 4. Overall Model Comparison

| Model               | Type              | Accuracy | Precision | Recall | F1-score |
|---------------------|-------------------|----------|-----------|--------|----------|
| Logistic Regression | Probabilistic     | 0.620    | 0.100     | 0.84   | 0.178    |
| Naïve Bayes         | Probabilistic     | 0.381    | 0.072     | 0.98   | 0.134    |
| SVM                 | Non-Probabilistic | 0.693    | 0.121     | 0.84   | 0.211    |
| KNN                 | Non-Probabilistic | 0.943    | 0.167     | 0.04   | 0.065    |

(All figures use each model’s final tuned version. Logistic Regression and SVM are also ranked by ROC-AUC where available, since AUC is threshold-independent and reflects the reliability of the underlying probability/decision-function ranking.)

#### 5. Discussion and Conclusion

##### Comparing the models

Raw accuracy is a misleading yardstick for this problem: KNN reaches 94.3% accuracy while catching only 4% of real stroke cases, because strokes make up a small minority of the dataset and

a model can score well simply by predicting “no stroke” almost everywhere. Once recall and AUC are weighted alongside accuracy, a clearer picture emerges. Logistic Regression and SVM (tuned) both reach a recall of 0.84 with broadly comparable precision and F1-scores, making them the two strongest performers overall. Naïve Bayes reaches the highest raw recall (0.98) but its accuracy (38%) and AUC (0.798) reveal that this is achieved by flagging almost every patient as at-risk rather than by genuine discrimination. KNN is clearly the weakest model: its distance-based mechanism has no way to compensate for class imbalance, and tuning could not meaningfully improve it.

## Best-performing model

The tuned **Logistic Regression** model is identified as the best-performing model overall. It matches SVM's recall (0.84) while achieving the highest ROC-AUC of all models (0.838), meaning it ranks patients by risk more reliably than any other model, at any threshold — not just the one used for the confusion matrix above. It also has a practical advantage the other models lack: its coefficients are directly interpretable, so clinicians can see which factors (e.g. age, average glucose level, hypertension) are driving a given risk prediction, which matters for trust and adoption in a healthcare setting. SVM (sigmoid kernel) is a close second, matching Logistic Regression on recall and F1, but its decision function is harder to interpret and does not natively output calibrated probabilities.

## Strengths and limitations of each technique

**Logistic Regression.** Fast to train, produces interpretable coefficients and calibrated probability estimates, and responded well to both class weighting and L1 regularisation. Its main limitation is the assumption of a linear relationship between features and the log-odds of the outcome, which can limit performance on more complex, non-linear patterns.

**Naïve Bayes.** Extremely fast and simple, and achieved the highest raw recall of any model. Its core limitation is the assumption that features are conditionally independent given the class — an assumption clinical features such as age, hypertension, and glucose level routinely violate — which here produced poor calibration and very low precision (over-flagging nearly everyone as at-risk).

**Support Vector Machine.** Performed strongly once tuned, handling the imbalance well via class weighting and benefiting from a well-chosen kernel. Its limitations are reduced interpretability (no direct feature-importance equivalent to Logistic Regression's coefficients), higher computational cost, and the fact that its raw output is a decision-function score rather than a genuine probability.

**K-Nearest Neighbours.** Simple and non-parametric, requiring no assumptions about the underlying data distribution. Its critical limitation for this task is that it has no mechanism to account for class imbalance: with strokes as a small minority, a new patient's nearest neighbours are overwhelmingly non-stroke cases regardless of k, weighting, or distance metric, making it unsuitable for this dataset as-is.

## Recommendation

For stroke risk screening on this dataset, the tuned **Logistic Regression** model is recommended as the most trustworthy option: it offers the best combination of recall and discriminative power (AUC), along with the interpretability that clinical decision-making needs. In practice, its probability output would allow clinicians to adjust the decision threshold to trade recall against false-alarm rate depending on available follow-up resources, rather than being locked into a single fixed cut-off.



Naïve Bayes should be avoided despite its high headline recall, since its predictions are not well-calibrated and would flood a clinical workflow with false positives.